

GANESH SENRAYAN

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RESEARCH INTERESTS

Machine Learning, Deep Learning, Large Language Models, LLM Agents, Knowledge Graphs, Retrieval-Augmented Generation, Multi-Armed Bandits

EDUCATION

Indian Institute of Technology Madras, Chennai, India
Integrated M.Tech. in Data Science & B.Tech. in Mechanical Engineering
Advisor: Prof. Balaraman Ravindran, DSAI Dept.
Nov 2020 — May 2025
Cumulative GPA: 8.67/10.0

Mahatma Montessori School, Madurai, India
Class XII (CBSE)
Jun 2019 — Apr 2020
Percentage: 94.20%

Vikaasa School, Madurai, India
Class X (ICSE)
Jun 2017 — Apr 2018
Percentage: 94.33%

PUBLICATIONS

With Centre for Responsible AI & Google DeepMind

- Parthasarathy, A., Subramanian, C., **Senrayan G.**, Adappanavar, S., Ravindran, B., Taneja, A., and Tambe, M. Multilinguality in LLM-Designed Reward Functions for Restless Bandits: Effects on Task Performance and Fairness. (*Deployable AI Workshop, AAAI 2025*).

With Fujitsu Research

- Agarwal, A., **Senrayan G.**, and Devaguptapu, C. Hybrid Graphs for LLM-based Question Answering on Table-and-Text Sources. (*NAACL 2025*)

RESEARCH EXPERIENCE

Fujitsu Research
AI Researcher
Bengaluru, India
Jul 2025 — Present

- Improving memory architectures for agentic LLM systems to show increased recall on long-horizon chat tasks
- Building a dataset to evaluate effectiveness of LLM-powered coding agents on long horizon tasks

Centre for Responsible AI, IITM
Student Researcher (Master's Thesis)
Chennai, India
Aug 2024 — May 2025
Guides: Prof. Balaraman Ravindran and Dr. Chandrasekhar Subramanian

- Collaborating with **Google DeepMind (Prof. Milind Tambe and Dr. Aparna Taneja)** to work on using LLM-generated reward functions for Restless Multi-Armed Bandits (RMABs) with applications in public health
- Analyzed impact of multilingualism, complexity and phrasing of the prompts on performance & fairness of the algorithm
- Investigating scalable and deployable solutions for making the RMAB algorithm effective at the grassroots level

Fujitsu Research
Research Intern
Bengaluru, India
May 2024 — Aug 2024
Guides: Ankush Agarwal, Pranoy Panda and Chaitanya Devaguptapu

- Developed a novel Hybrid Graph-based approach for complex question answering using LLMs over table and text sources
- Achieved an increase in performance metrics such as Exact-Match by **9.5%**, F1 score by **8%**, reduction in token size & cost by **44%** over SoTA methods in Hybrid-QA and OTT-QA datasets
- Engineered Agentic Architecture for RAG using Knowledge Graphs and deployed a working demo using FastAPI, Next.js

ENGINEERING EXPERIENCE

Sarvam.ai
AI Engineer Intern
Chennai, India
Feb 2024 — Mar 2024

- Benchmarked the ability of more than 5 LLMs to call tools accurately on the Berkeley Function Calling Leaderboard
- Experimented with the ReAct framework of prompting and achieved a **2%** increase in the Overall Accuracy category
- Built a web-app PoC using Generative UI created by LLM Agents for booking flights through voice commands

- Developed a Python pipeline to transliterate Alteryx XML files and generate equivalent Informatica XML files
- Deployed a directed graph to replicate ETL workflows, optimizing migration with graph traversal algorithms
- Automated the migration of more than **80** ETL workflows, reducing manual intervention by more than **165** man-hours

KEY PROJECTS

Generative Models for Clustering single-cell RNA-sequencing Data

Guide: Prof. Manikandan Narayanan

- Experimented with Variational Auto Encoders for clustering single-cell RNA sequencing data to identify novel cell types
- Implemented state-of-the-art papers such as scAce and scVGAE and performed a comparative study of the methods
- Proposed a novel solution combining Variational Auto Encoders and PhenoGraph, a graph-based clustering algorithm, achieving an increase in ARI by **2%**

Trustpilot Review Analysis

Guide: Dr. Sudharsun Sandhiappan

- Developed a 2-step data analytics pipeline for analysing trust pilot reviews and parallelized it using Apache Beam
- Used a simple RNN model for review sentiment prediction and orchestrated the entire pipeline using Airflow
- Served the model through FastAPI, containerized it using Docker and monitored it through Prometheus and Grafana

Image Captioning

Guide: Prof. Chandra Sekhar C

- Built an Image Captioning module using CNN and NetVLAD as the encoder and LSTM as the decoder
- Trained the module on the COCO Captions dataset, achieving a BLEU-1 score of **0.74**, BLEU-4 score of **0.28**
- Implemented various state-of-the-art techniques such as Meshed-Memory Transformers, ExpansionNet, etc.

SELECTED COURSES

- Pattern Recognition and Machine Learning
- Fundamentals of Deep Learning
- Algorithmic Approaches in Computational Biology
- Reinforcement Learning
- Software Engineering
- Mathematical Foundations of Data Science

EXTRA-CURRICULAR ACTIVITIES

Hues of India

Volunteer

Chennai, India
Feb 2024 — May 2024

- Volunteered at Hues of India, an NGO dedicated to improving lives of students through therapeutic art sessions
- Built a mobile application using Next.js, Django and AWS to improve participant engagement by gamifying the process

InstiSpace

Project Member

Chennai, India
Dec 2022 — Apr 2023

- Developed InstiSpace, a mobile application for students of IIT Madras with more than 5k downloads in the Play Store
- Used a tech stack consisting of Flutter for the frontend, NestJS with GraphQL for the backend and AWS for deployment

Programming Club IITM

Co-ordinator

Chennai, India
Apr 2021 — May 2022

- Coordinated a Summer Coding Camp, leading sessions on Data Structures and Algorithms for more than 1,500 students
- Organized competitive programming contests including Reverse Coding, Code Golf, with more than 2,000 participants

SKILLS

- **Programming Languages:** Python, C++, JavaScript, TypeScript, Dart
- **Libraries and Packages:** Pytorch, Tensorflow, Langchain, Langgraph, Sci-kit learn, NumPy, Pandas, Next.js, FastAPI
- **Tools and Software:** Docker, Apache Beam, Airflow, Prometheus, L^AT_EX